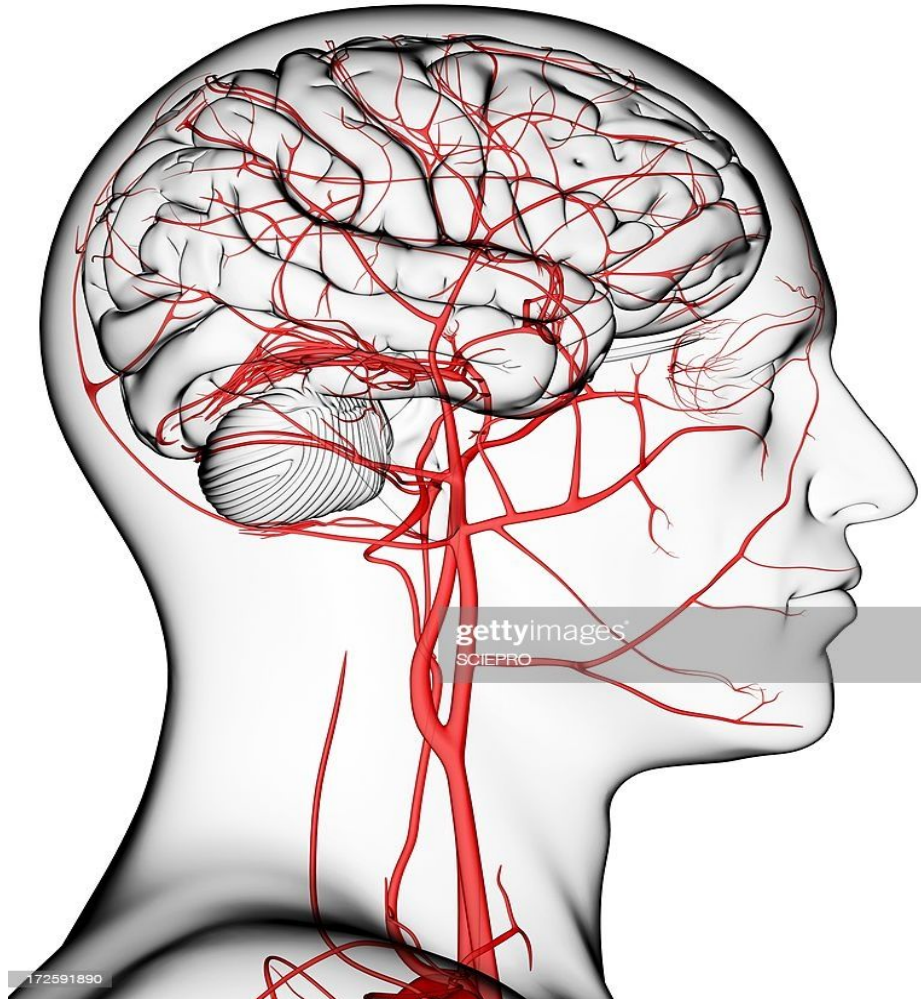


Blood Supply of Brain



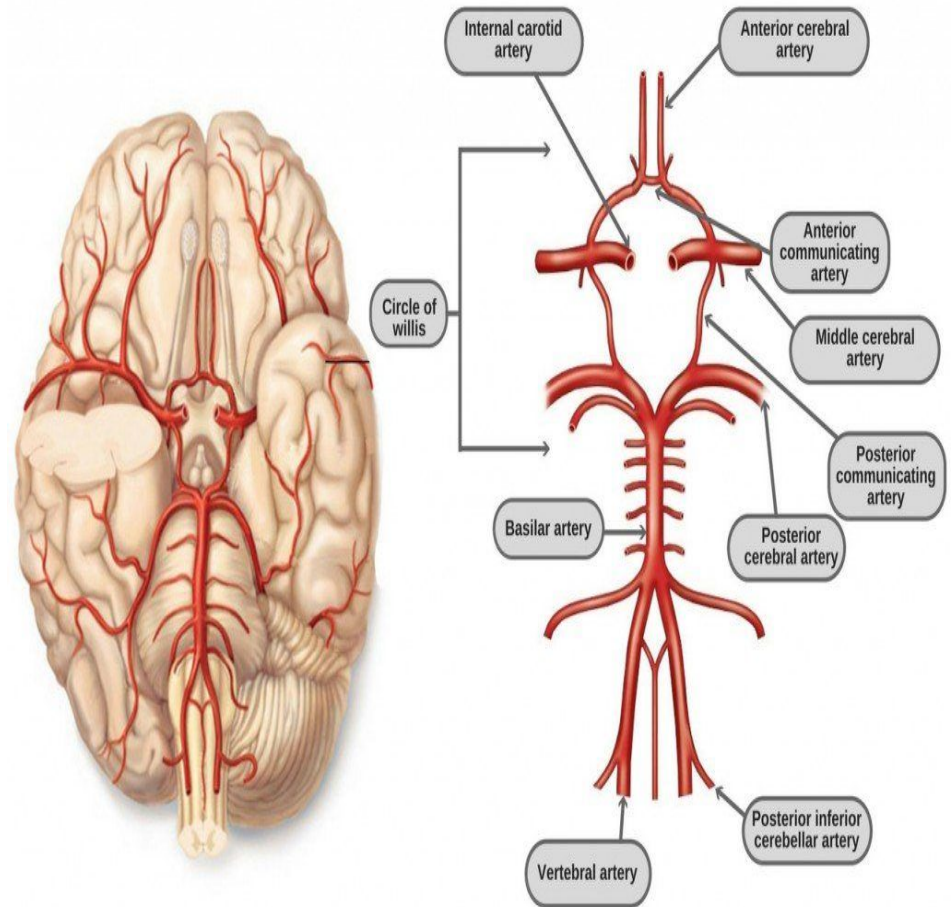
Learning objectives:

At the end of the lecture, the students should be able to:

- **Enlist the main arteries supplying the brain.**
- **Describe the formation of arterial circle of Willis.**
- **Enlist the branches of internal carotid, vertebral and basilar arteries.**
- **Describe the distribution of branches of internal carotid, vertebral and basilar arteries.**

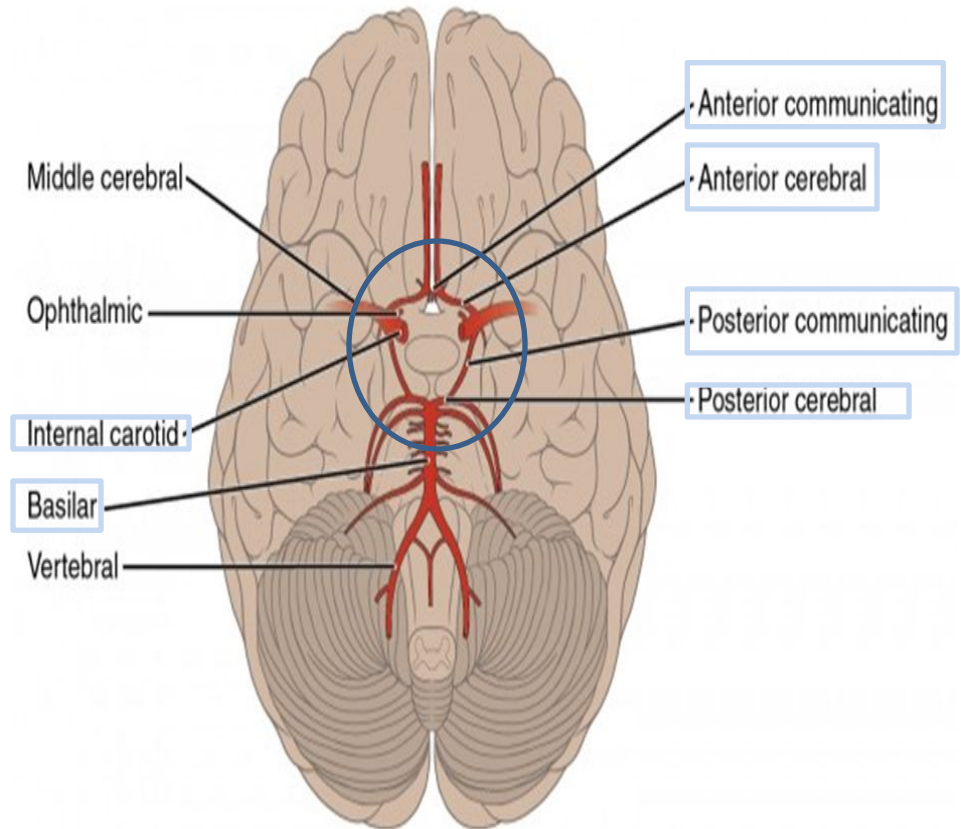
Arteries of Brain:

- Brain is supplied by two **internal carotid** and two **vertebral arteries**.
- The arteries lie in the subarachnoid space.
- Their branches anastomose on the inferior surface of brain to form circle of Willis.



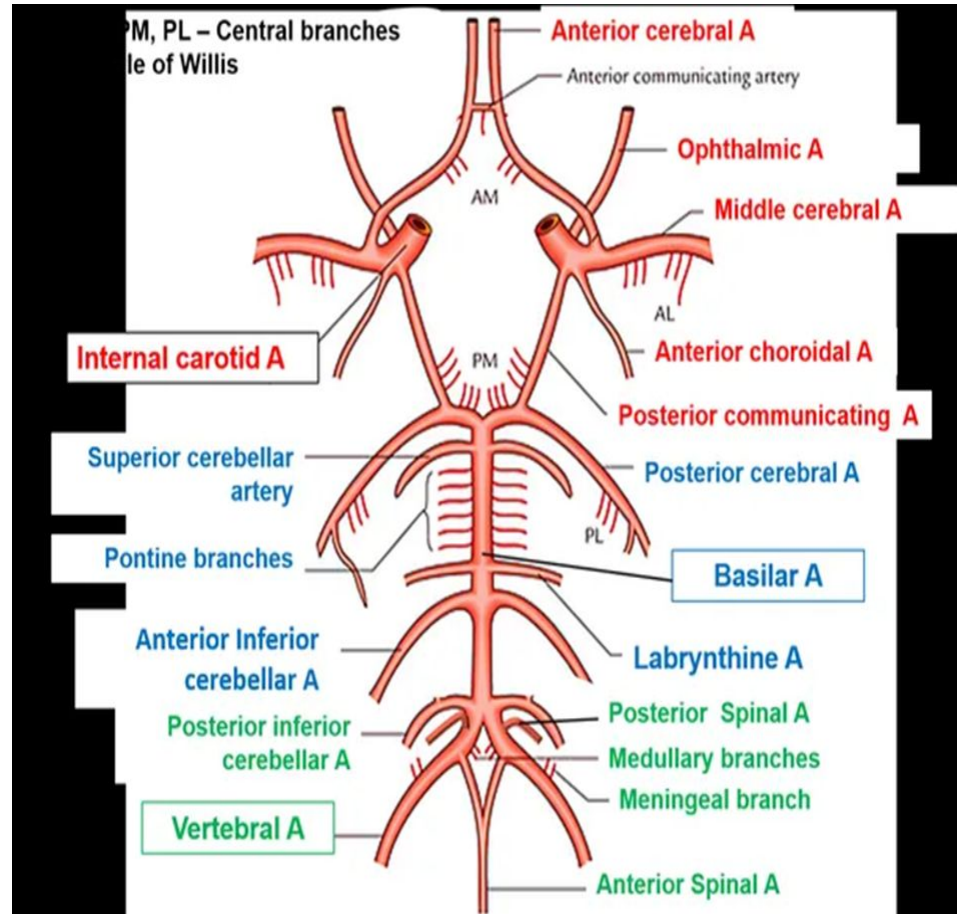
Circle of Willis:

- Lies in the interpeduncular fossa at the base of brain.
- Formed by anastomosis between two **internal carotid** and two **vertebral arteries**.
- **Contributing arteries:** anterior communicating, anterior cerebral, internal carotid, posterior communicating, posterior cerebral and basilar arteries.
- Circle of Willis allows blood coming from internal carotid or vertebral arteries to be distributed to any part of cerebral hemisphere.
- Cortical and central branches arise from circle and supply brain substance.



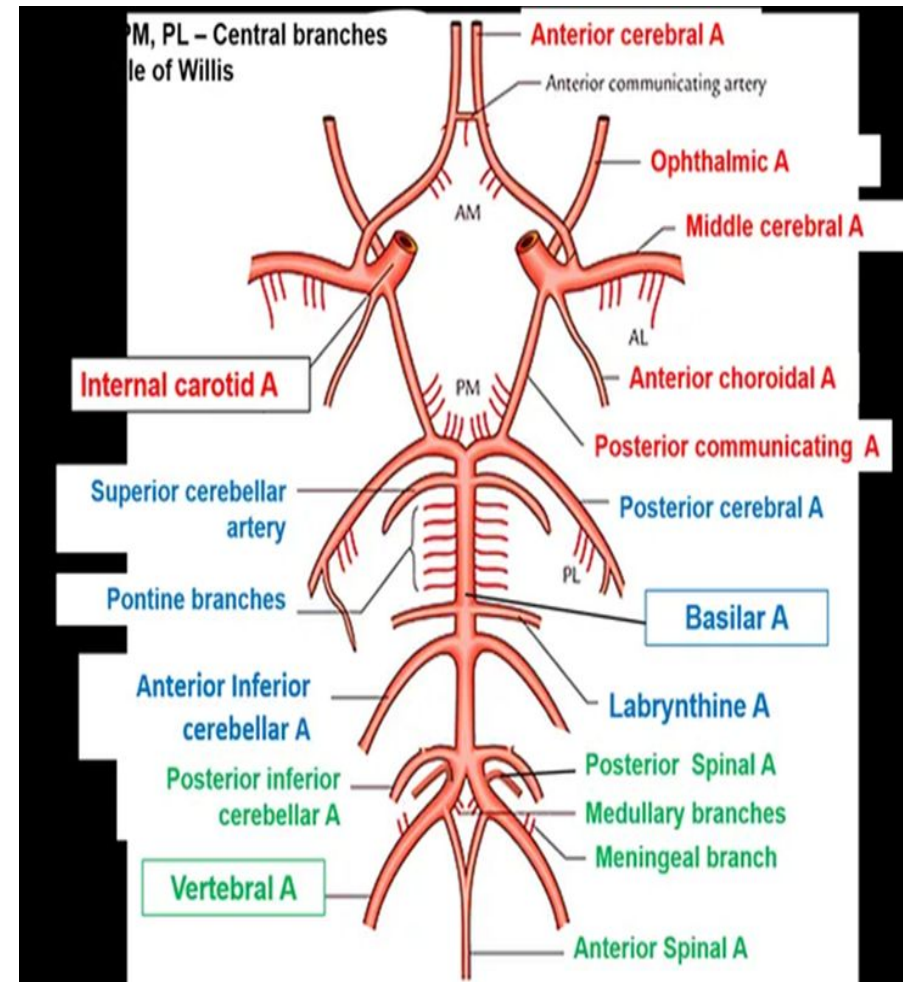
Internal carotid artery:

- Begins at bifurcation of common carotid artery.
- Ascends the neck and enters cranial cavity by passing through carotid canal in temporal bone.
- Runs horizontally forward through cavernous sinus.
- Enters subarachnoid space by piercing dura matter.
- Ends by dividing into anterior and middle cerebral arteries.



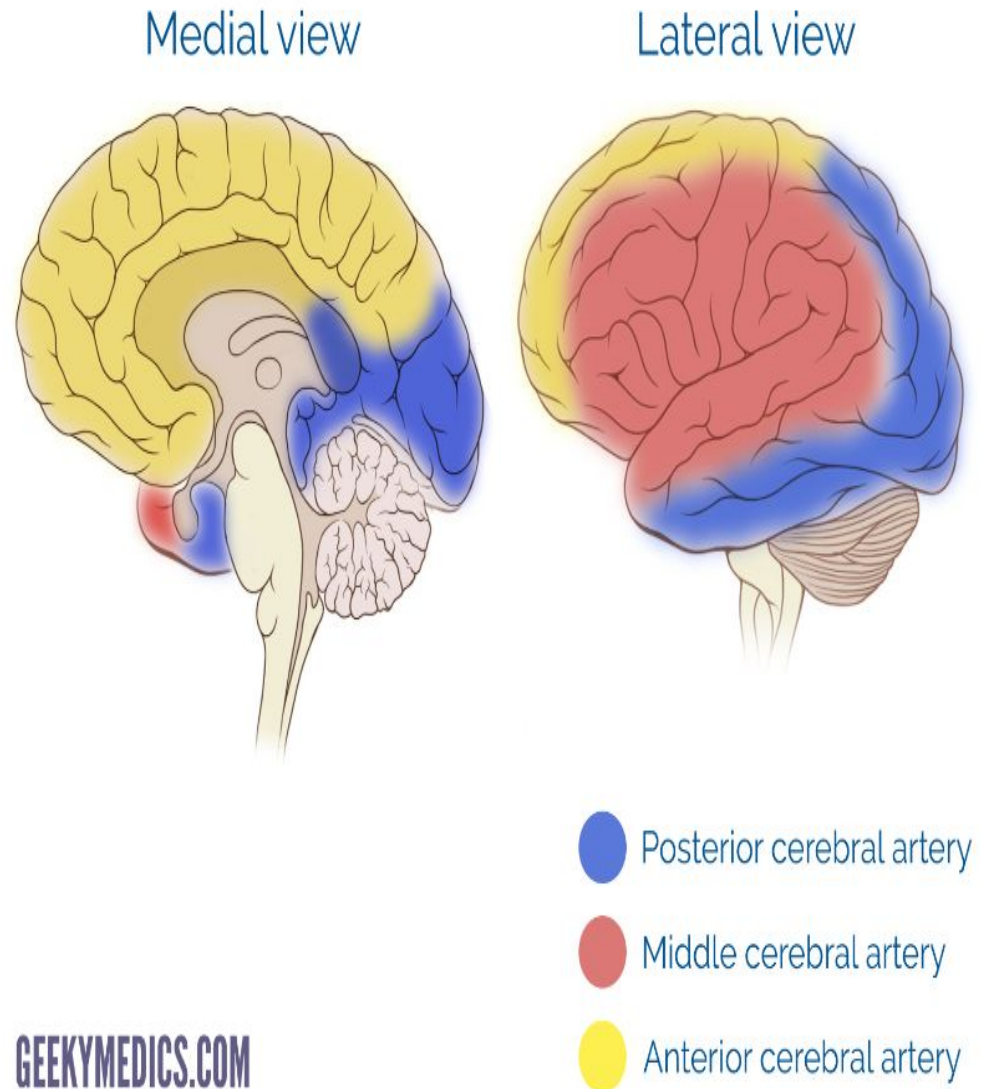
Branches of internal carotid artery and their distribution:

- I. **Ophthalmic artery:** supplies eye and other orbital structures.
- II. **Posterior communicating artery:** joins posterior cerebral artery forming a part of circle of Willis.
- III. **Choroidal artery:** ends in choroid plexus of lateral ventricle and supplies crus cerebri, lateral geniculate body etc.
- IV. **Anterior cerebral artery:** is a smaller terminal branch of internal carotid artery.
 - In the longitudinal fissure of cerebrum it is joined by anterior cerebral artery of opposite side by anterior communicating artery.
 - It curves backward over corpus callosum and finally anastomoses with posterior cerebral artery.



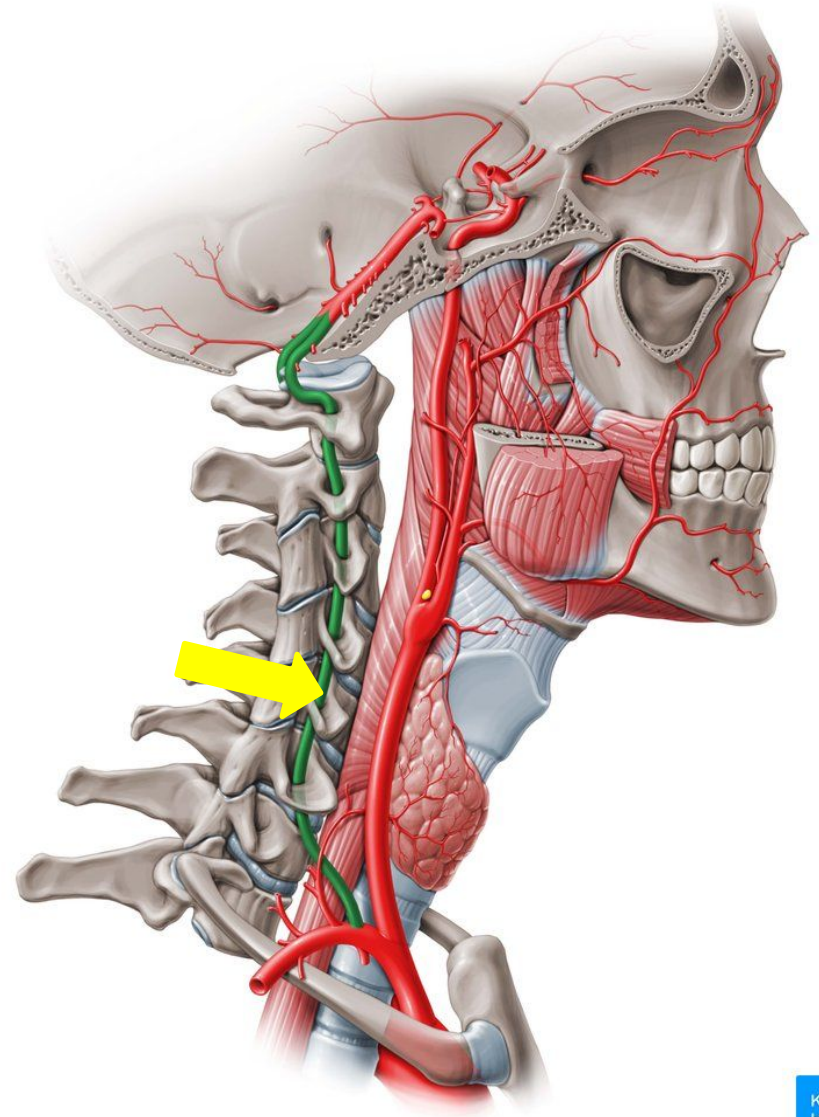
- **Distribution of Anterior cerebral artery:** cortical branches supply all the **medial surface of cerebral cortex upto parieto-occipital sulcus**.
- Also supplies a **strip of cortex 1 inch wide on the adjoining lateral surface**, thus it **supplies 'leg area' of precentral gyrus**.

- V. **Middle cerebral artery:** largest branch of internal carotid which runs laterally in lateral sulcus.
- **Distribution:** supplies entire lateral surface of cerebral hemisphere except for the narrow strip (leg area), occipital pole and inferolateral surface of hemisphere.
 - Thus the artery supplies all motor area except leg area.



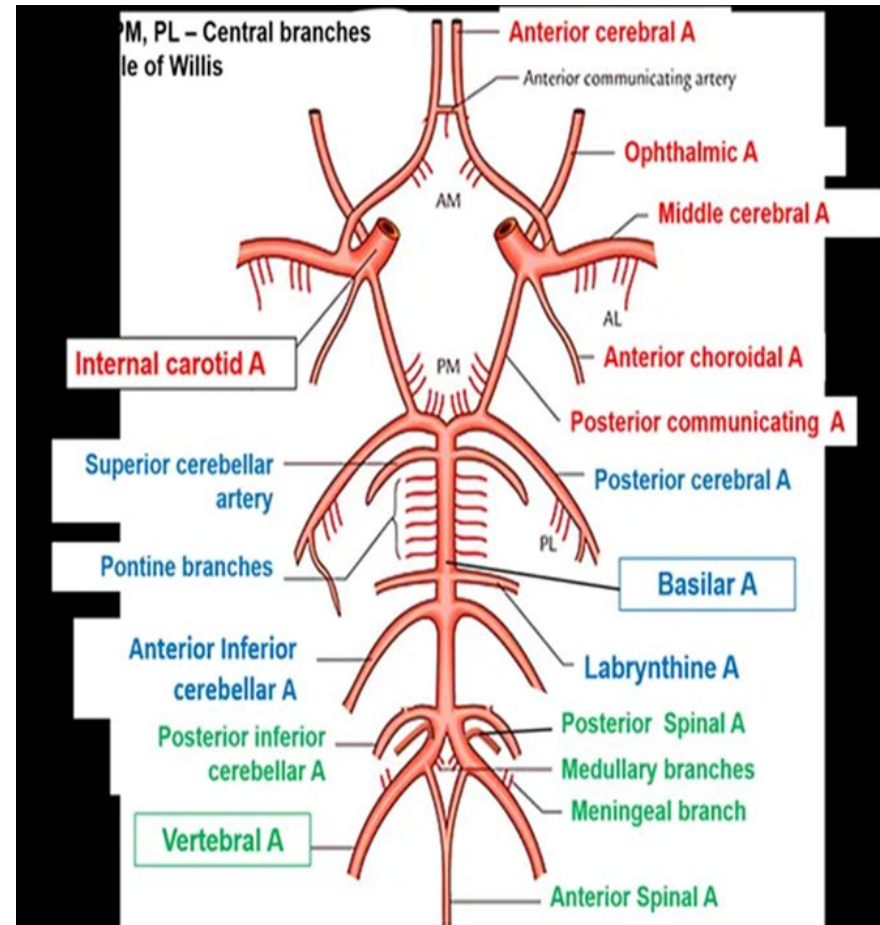
Vertebral artery:

- It is a branch of first part of subclavian artery.
- Ascends neck through transverse foramina of upper six cervical vertebrae.
- Enters skull through foramen magnum and pierces dura matter to enter subarachnoid space.
- Then passes medially and upward on medulla to reach lower border of pons.
- Here it joins the vessel of opposite side to form basilar artery.



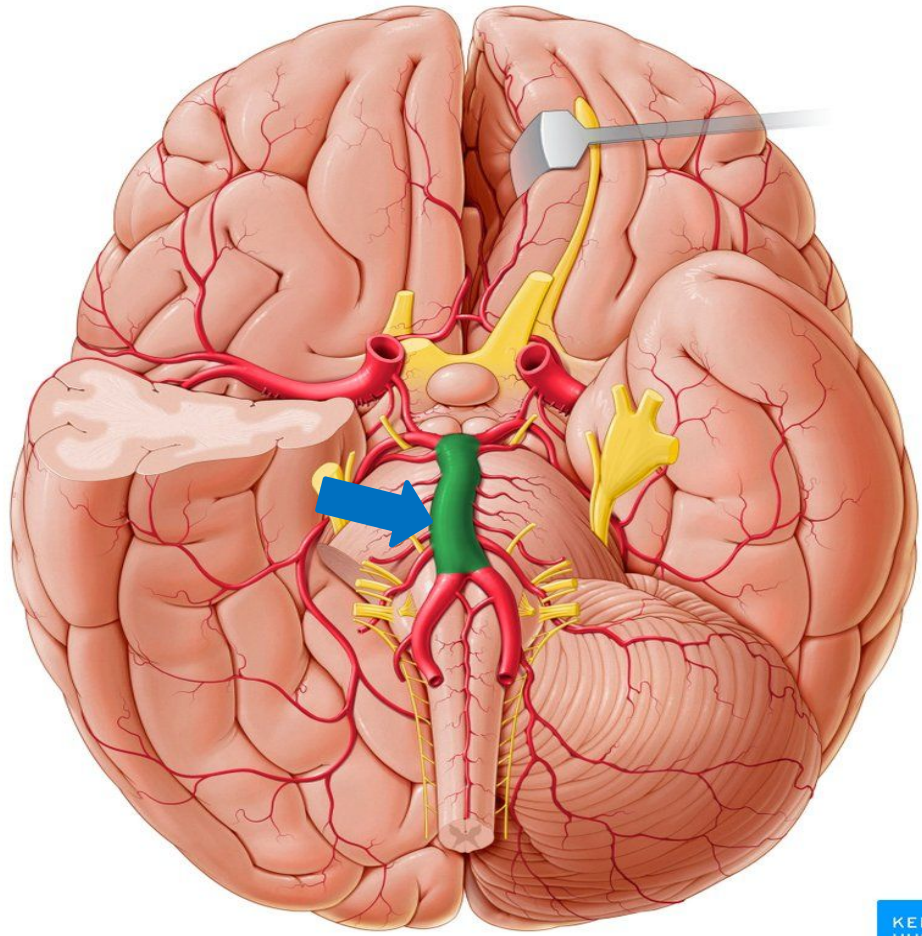
IV. Posterior inferior cerebellar artery:
supplies inferior surface of vermis, central nuclei of cerebellum and undersurface of cerebellar hemisphere.

V. Medullary arteries:
supply medulla oblongata.



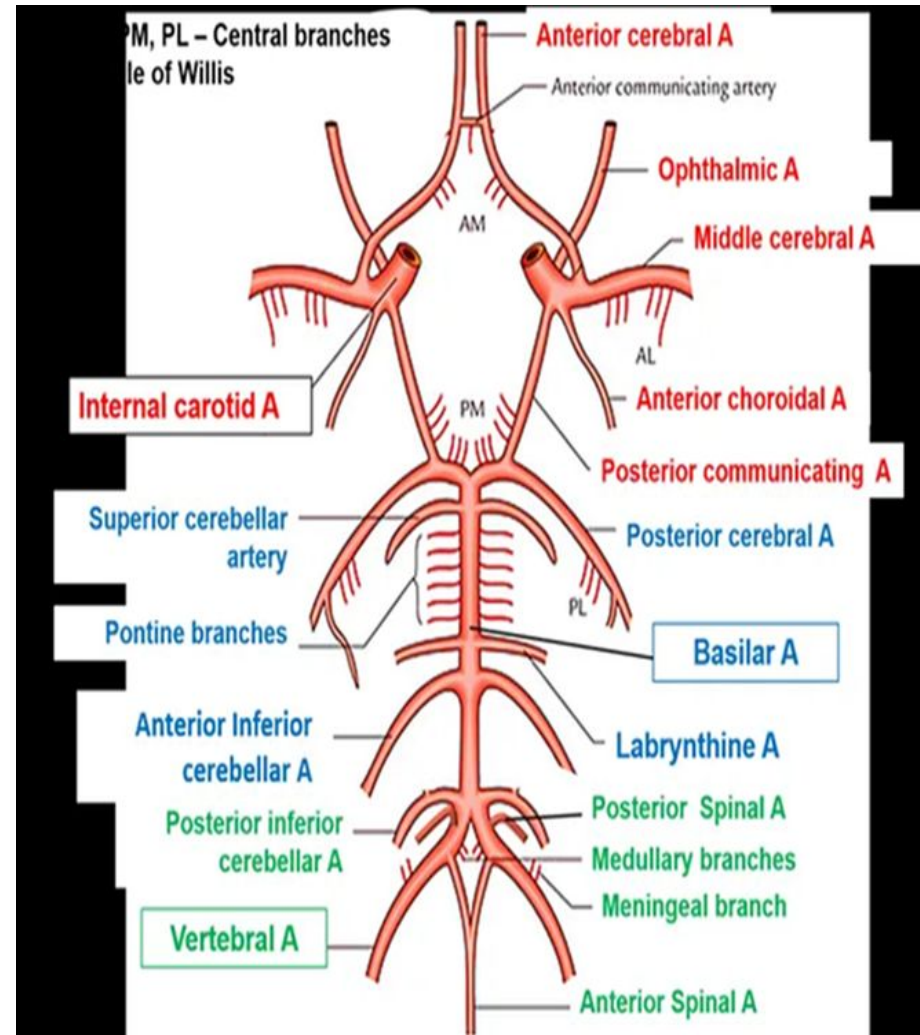
Basilar Artery:

- Formed by the union of two vertebral arteries.
- Ascends in the basilar groove on ventral surface of pons.
- At the upper border of pons, it divides into two posterior cerebral arteries.



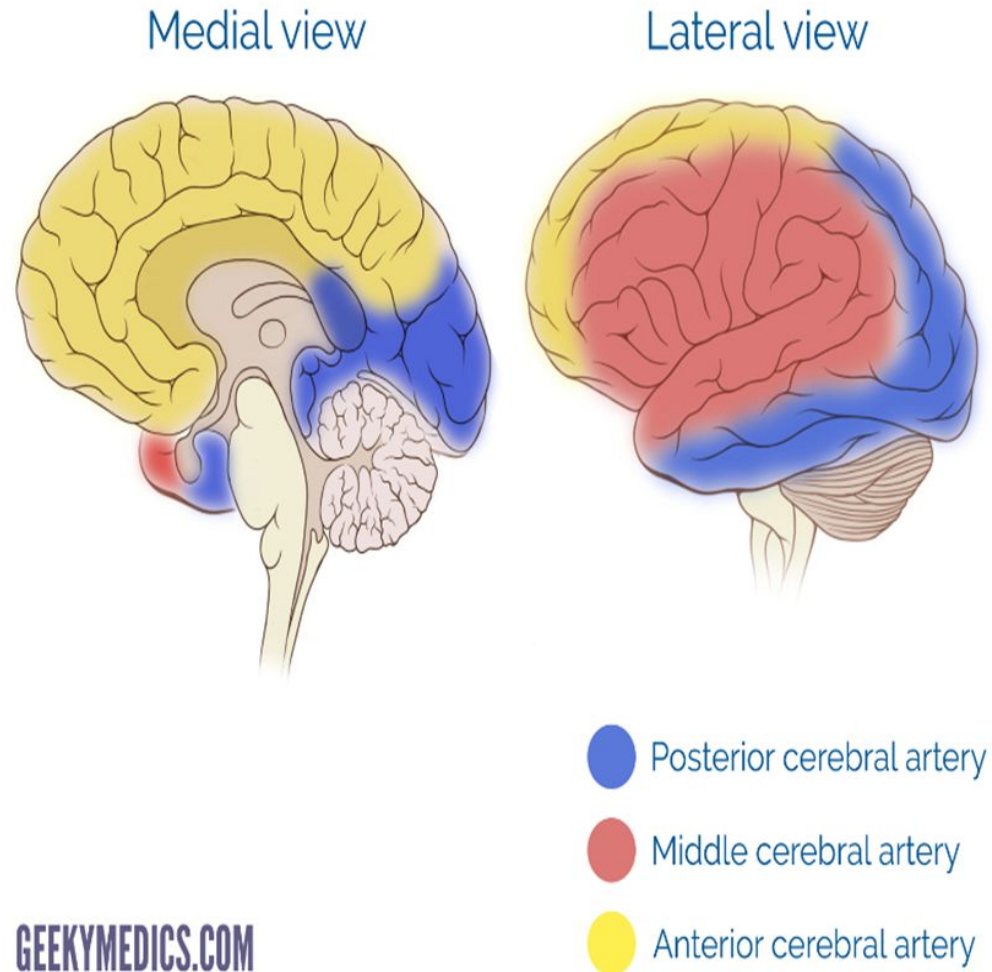
Branches of basilar artery and their distribution:

- I. **Pontine arteries:** supply pons.
- II. **Labyrinthine artery:** supplies internal ear.
- III. **Anterior inferior cerebellar artery:** supplies anterior and inferior parts of cerebellum.
- IV. **Superior cerebellar artery:** supplies superior surface of cerebellum.



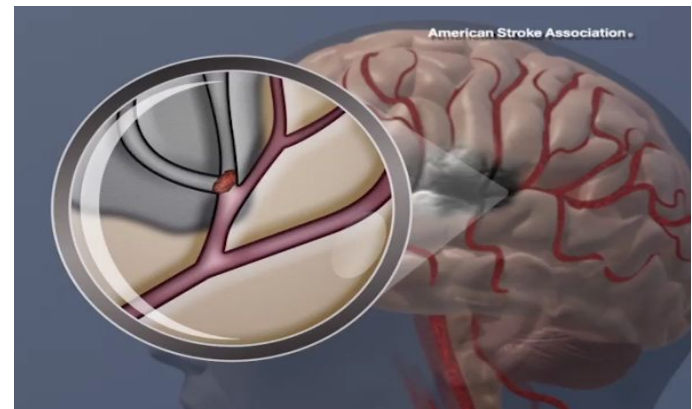
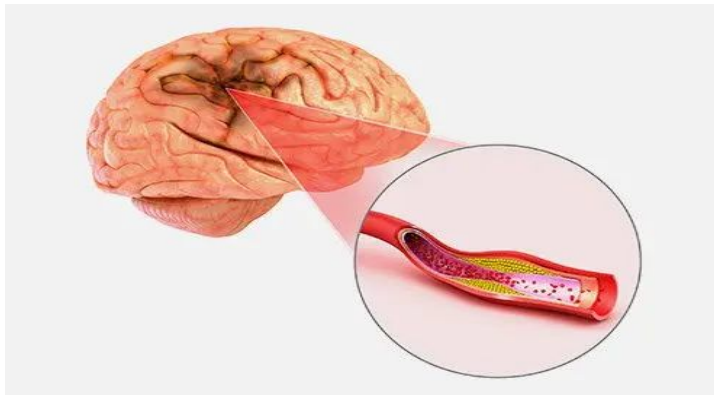
V. Posterior cerebral artery:
curves laterally and backward
around midbrain.

- Joined by posterior communicating artery.
- **Distribution:** cortical branches supply inferolateral and medial surface of temporal lobe and lateral and medial surfaces of occipital lobe, thus it supplies visual cortex.
- **Central branches:** supply parts of thalamus, midbrain, medial geniculate bodies etc.
- **A choroidal branch:** supplies choroid plexus of lateral and third ventricles.



Clinical Anatomy:

- The brain receives about 15% of the resting cardiac output.
- Unconsciousness happens in 5 to 10 seconds if blood supply to is completely ceased.
- Neuronal function stops after about 1 minute & irreversible changes start taking place after about 4 minutes following the arrest of cerebral blood flow.

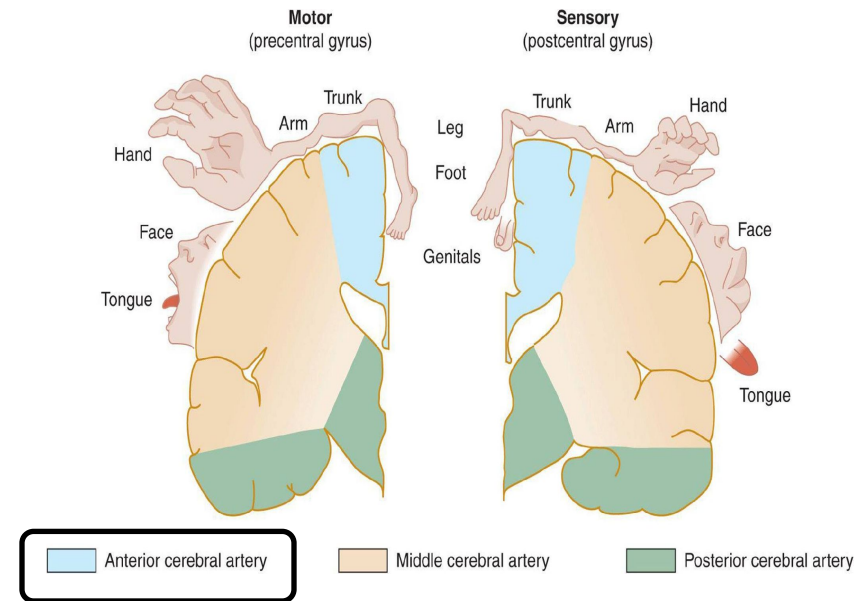


Cerebral Artery Syndromes:

Anterior Cerebral Artery Occlusion

Blockage of anterior cerebral artery (distal to anterior communicating artery) may lead to:

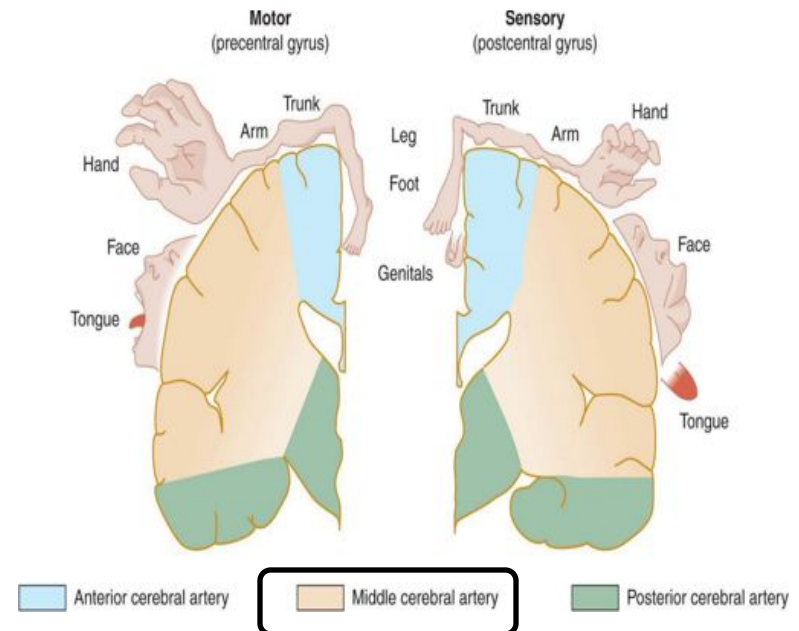
- Contralateral hemiparesis & hemisensory loss ***mainly affecting leg & foot*** (paracentral lobule).
- Personality changes, apathy, & impairment to identify objects (frontal & parietal lobe), &
- Urinary incontinence.



Middle Cerebral Artery Occlusion

Blockage of middle cerebral artery may produce following signs & symptoms:

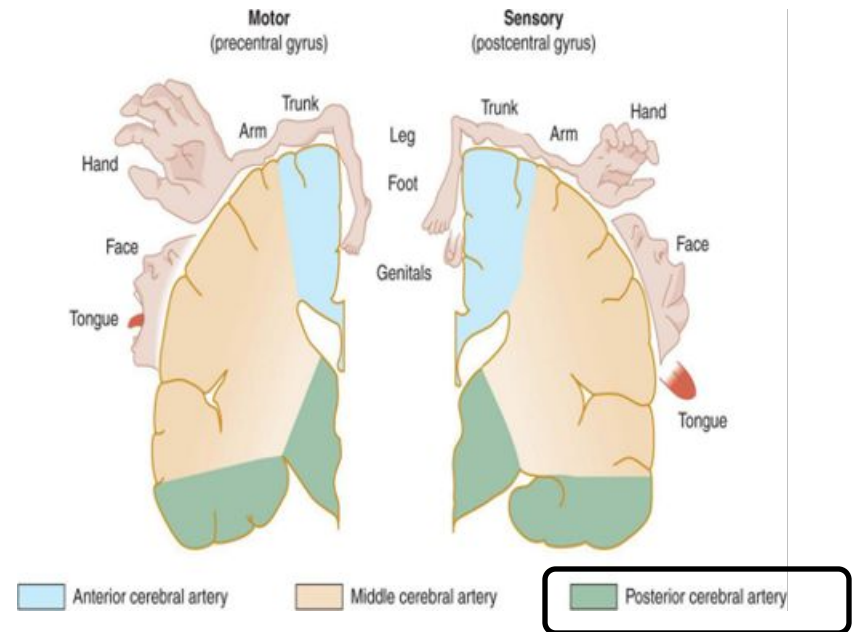
- Contralateral hemiparesis & hemisensory loss **mainly face & arm** (precentral & postcentral gyri).
- Contralateral homonymous hemianopia (affecting optic radiation)
- Aphasia (mostly in case of left hemisphere)
- Anosognosia (mostly involving right hemisphere)



Posterior Cerebral Artery Occlusion

Blockage of posterior cerebral artery may produce following signs & symptoms:

- Contralateral homonymous hemianopia with macular sparing
- Visual agnosia
- Memory impairment.



Internal Carotid Artery Occlusion

Internal carotid artery occlusion may lead to:

- The signs & symptoms similar to those of middle cerebral artery including contralateral hemiparesis & hemianesthesia.
- Partial or complete loss of vision on the same side.

Stroke Syndromes

★ Middle Cerebral Artery — “MCA = Most Common Arm”

- MCA → Arm > leg weakness
- Aphasia if left side
- Hemineglect if right side



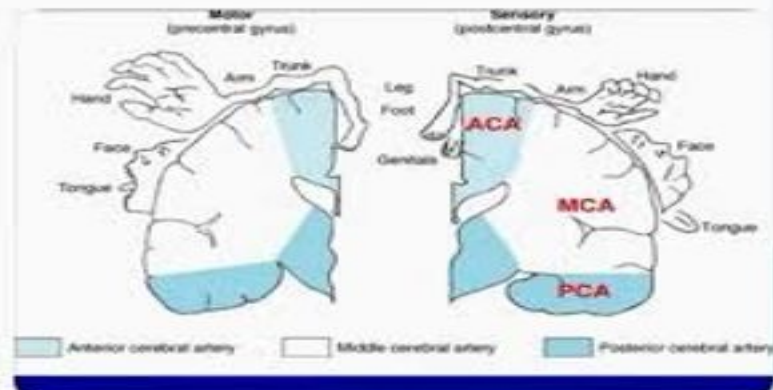
★ Anterior Cerebral Artery — “ACA = Affects the Ankle”

- Leg > arm weakness
- Personality change



★ Posterior Cerebral Artery — “PCA = Picture Problems”

- Visual field defects
- Alexia without agraphia



References:

- **CLINICAL NEUROANATOMY**

Richard S. Snell

Chapter # 17

8th edition

Pages # 475 - 480